

KUSETTI TARUN

☎ 8688014671

✉ tarunkusetti4002@gmail.com

📍 Visakhapatnam

🐙 GitHub

🌐 LinkedIn

Summary

Electronics and Communication Engineering student with hands-on knowledge of C programming, Embedded Systems, and STM32 microcontrollers. Skilled in interfacing and programming STM32 peripherals, sensor integration, and IoT-based project development. Capable of designing, debugging, and implementing real-time embedded applications with a strong interest in embedded firmware development, automation, and hardware-software integration.

Education

Bachelor of Technologies (ECE), Dr. Lankapalli Bullayya College of Engineering 2022 – 2026
CGPA: 8.04% Visakhapatnam

Intermediate (MPC), Gnanajyothi Junior College 2020 – 2022
Percentage: 87.6% Srikakulam

Gayatri High School 2020
Percentage: 95% Srikakulam

Skills

- **Languages:** C, Embedded C
- **Microcontrollers:** STM32, Arduino, ESP32
- **Embedded Concepts:** GPIO, UART, I2C, SPI, Timers, ADC, PWM, Peripheral Interfacing
- **Tools & IDEs:** STM32CubeIDE, Arduino IDE, KiCad
- **Operating System:** Linux (Ubuntu)
- **Cloud Platform:** Rightech Cloud
- **Embedded Software:** Linux Internals, Embedded Linux(Rugged Board), Zephyr RTOS, Device Drivers

Internships

Industrial Training – BHEL (Bharat Heavy Electricals Limited)

Duration: 1 Month

Gained basic understanding of CNC machinery and its applications in industrial manufacturing. Observed real-time automation processes and learned about production workflows and safety standards.

Certificates

- Industrial Training at Bharat Heavy Electricals Limited.
- NPTEL - Micro Sensors and Nano Sensors

Projects

Autonomous Mobile Robot Based IOT Integration

Technologies: STM32CubeIDE, Embedded C, UART, PWM, Wi-Fi, IoT.

- Developed an IoT-based autonomous surveillance robot using **STM32F446RE** for real-time monitoring and control.
- Integrated **ultrasonic and flame sensors** using GPIO for obstacle detection and hazard identification.
- Configured and interfaced **ESP32-CAM** module using UART for wireless video streaming.
- Implemented **Wi-Fi-based communication** for remote monitoring and control.
- Controlled robot movement using **L298N motor driver** and PWM signals.
- Enabled **servo motor-based camera positioning** and **buzzer alerts** for safety indication.

IoT-Based Smart Parking System

Technologies: STM32CubeIDE, Embedded C, UART, SPI, I2C, PWM, HTTPS/REST, IoT, Cloud Database, Frontend

- Developed an IoT-based Project using STM32F446RE with RFID-based access and automated gate control.
- Interfaced peripherals using SPI, I2C, GPIO, and PWM, and integrated a Wi-Fi module over UART for secure HTTPS/REST communication with a cloud database.
- Implemented RFID authentication, parking fee calculation, real-time parking status updates, and reliable synchronization after system resets.
- Designed a web-based admin dashboard to monitor parking status, slot availability, registered users, and parking earnings in real time.

Environmental Monitoring System using Zephyr RTOS

- Technologies: Zephyr RTOS, Embedded C, STM32 Nucleo-F446RE, BME280, I2C, UART, WE-10, RightTech IoT Cloud
- Developed an environmental monitoring application using Zephyr RTOS on the STM32 Nucleo-F446RE board by interfacing the BME280 sensor over I2C.
- Configured hardware using DeviceTree overlays and implemented sensor data acquisition using Zephyr device drivers.

Transmitted real-time temperature, humidity, and pressure data to RightTech IoT Cloud through the WE-10 Wi-Fi module using UART.

Traffic Light Control System

Technologies: STM32CubeIDE, Embedded C, GPIO, Timers, UART

- Designed and developed an automated traffic light control system using the **STM32F446RE** microcontroller.
- Implemented a **Finite State Machine (FSM)** to control Red, Yellow, and Green signal transitions with predefined timing.
- Configured **hardware timers** to generate accurate one-second intervals for traffic signal operation.
- Interfaced a **7-segment display** to provide a real-time countdown for each traffic signal state.
- Utilized **GPIO** to control LEDs representing traffic signals and drive the display interface.
- Developed modular firmware using the **STM32 HAL** library, improving code readability, scalability, and maintainability.
- Verified signal timing accuracy and state transitions through testing and debugging.

PERSONAL DETAILS

Father's Name: K. Chinna Rao

Mother's Name: K. Parvathi

Date of Birth: 11/03/2004

Address: kotha Peta, Kotabommali Mandal, Srikakulam District, Andhra Pradesh